

BSE Group Assignments: Task Sheet

Mathematics for Machine Learning, Formative 2 PCA, Cohort 2, Team 15

Team Members

Team Members	Role / Responsibility	Email
Teta Butera Nelly	Dataset, data handling, PCA code, plots, benchmarks, GitHub repo	t.nelly@alustudent.com
Mutoni Keira	Written answers, Colab check, GitHub README and data dictionary	m.keira@alustudent.com

Task Allocation and Tracker

Task Allocated	Assigned Member	Deadline	Completion Status	Reviewed by Team?	Comments
Picked the dataset: Our World in Data energy data for 54 African countries (2000-2022), and checked it against the brief	Teta Butera Nelly	2026-09-30	Completed	Yes	944 real blank cells in the raw file, 19 features in the end.
Data handling: found the blanks, filled them in, turned the text columns into numbers and logged the skewed ones	Teta Butera Nelly	2026-09-30	Completed	Yes	Explained step by step in the notebook.
PCA from scratch in numpy: standardize, covariance, eigendecomposition, sort and project (Steps 1-7), with a check after each step	Teta Butera Nelly	2026-09-30	Completed	Yes	Eigenvalues add up to exactly 19 and the eigenvectors are orthonormal.
Worked out how many components to keep from the explained variance, with scree and cumulative variance plots (Task 2 code)	Teta Butera Nelly	2026-09-30	Completed	Yes	7 components keep 90% of the variance.
Before and after PCA plots, component loadings and a chart of what information is lost per feature (Step 8)	Teta Butera Nelly	2026-09-30	Completed	Yes	
Speed tests for big data: loops vs vectorized, float32 vs float64, chunked covariance and SVD, up to 1,000,000 rows (Task 3)	Teta Butera Nelly	2026-09-30	Completed	Yes	
Wrote the covariance explanation and the three interpretation answers (before/after plot, number of	Mutoni Keira	2026-09-30	Completed	Yes	In the notebook and in the Google Doc.

components and trade-offs, what is lost)					
Ran the notebook in Google Colab and checked every output, then post a short note on GitHub saying it works	Mutoni Keira	2026-09-30	In progress	No	Ran it in Colab with no errors. Still need to post the GitHub note.
GitHub: accepted the invite, worked on a branch and opened a pull request for the README and data dictionary	Mutoni Keira	2026-09-30	Completed	Yes	Pull request #1 is merged.
GitHub: added a how-to-run-in-Colab section and a results summary to the README	Mutoni Keira	2026-09-30	Completed	Yes	
GitHub: wrote data/DATA_DICTIONARY.md for the 15 features and the sub_region column	Mutoni Keira	2026-09-30	Completed	Yes	
Last steps: download the Google Doc as a PDF, merge it with the task sheet PDF, final check and submit on Canvas	Teta Butera Nelly and Mutoni Keira	2026-09-30	In progress	No	Last step.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
2026-09-28	Work distribution	Teta Butera Nelly	Teta Butera Nelly and Mutoni Keira	Went through the brief and the rubric together and split the work. Nelly does the data, the PCA code, the plots and the benchmarks. Mutoni does the written answers, the Colab check, and the README and data dictionary.	Nelly finds a dataset and builds the notebook. Mutoni starts her parts once the notebook is ready.
2026-09-30	Execution	Teta Butera Nelly	Teta Butera Nelly and Mutoni Keira	The notebook, repo and report are done. Mutoni added her answers, the README and the data dictionary, and ran the notebook in Colab. We went through what's left: the task sheet, the final PDF and submitting.	Fill in the task sheet, download the Doc as a PDF, merge it with the task sheet PDF and submit on Canvas.